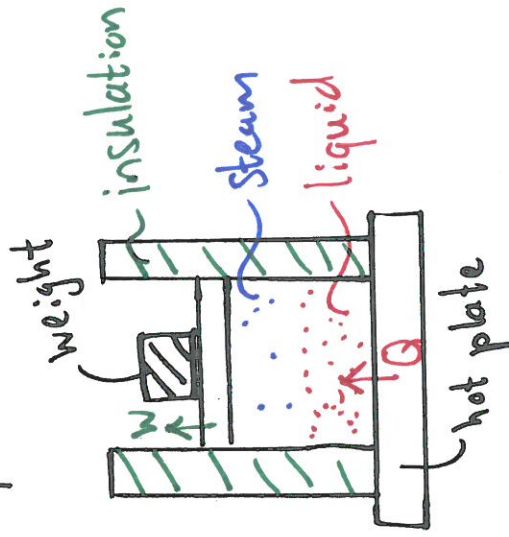


Example Let 1.00 kg of liquid water be at 100°C be converted to steam at 100°C by boiling at standard atmospheric pressure ($1.00 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$) in an arrangement as shown. The volume of water changes from $1.00 \times 10^{-3} \text{ m}^3$ as a liquid to 1.671 m^3 as steam.



(a) How much work is done on the system during this process?

$$W = - \int_{V_i}^{V_f} P dV \quad (\text{assume } P \text{ is constant})$$

$$= -P \int_{V_i}^{V_f} dV$$

$$= -P (V_f - V_i)$$

$$= - (1.01 \times 10^5 \text{ Pa} \times \frac{1 \text{ N/m}^2}{1 \text{ Pa}}) (1.671 - 10^{-3}) \text{ m}^3$$

$$= - 169 \times 10^3 \text{ N} \cdot \text{m} = - 169 \text{ J} = - 169 \text{ kJ}$$

$W < 0$ work done by gas on piston

(b) How much energy is transferred as heat during the process?

$$\begin{aligned} Q &= L_v \Delta m \\ &= (2256 \text{ kJ/kg})(1.0 \text{ kg}) \\ Q &= 2256 \text{ kJ} \end{aligned}$$

(c) What is the change in the internal energy of the system?

$$\begin{aligned} \Delta E_{\text{int}} &= Q + W \\ &= (2256 - 169) \text{ kJ} \\ &= 2087 \text{ kJ} \\ \Delta E_{\text{int}} &= 2.09 \text{ MJ} > 0 \quad (\text{internal energy increases}) \end{aligned}$$